

Stats 596 - Notes

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Contents

1	Chapter 1	2
1.1	Lecture 1: 9/1	2
1.2	Lecture 2: 9/3	2
1.3	Lecture 3: 9/10	4

Preface & Comments

1 Chapter 1

1.1 Lecture 1: 9/1

Remark (Data of Model). $(X_i, Y_i), i = 1, 2, \dots, n$ X_i is the covariate, Y_i is the response.

$$Y_i = X_i^T \beta^* + \epsilon_i, i = 1, 2, \dots, n$$

where $\beta^* \in \mathbb{R}^p$ is an unknown coefficient vector. and ϵ_i is the noise term.

Definition (Fixed Design.). $X_1, X_2, \dots, X_n$ are fixed and known and $Y_1, Y_2, \dots, Y_n$ are random. More specifically Y_i are independent and $E[Y_i] = X_i^T \beta^*$

This is the classical linear regression model. And ignore the randomness in X_i 's by conditioning some on X_i 's.

Most of the theory of historical linear regression is based on the fixed design assumption.

Definition (Random Design). $(X_i, Y_i), i = 1, 2, \dots, n$ are independent and identically distributed (i.i.d.) random variables, note they are pairs of random variables.

$$Y_i | X_i \sim N(X_i^T \beta^*, \sigma^2)$$

$$\epsilon_i | X_i \sim N(0, \sigma^2)$$

where $\epsilon_i = Y_i - X_i^T \beta^*$ is the noise term. More specifically $(X_i, Y_i) \sim P_{XY}$, where P_{XY} is the joint distribution of (X, Y) and $E[Y|X] = X^T \beta^*$

$E[Y|X] = X^T \beta^*$ is the conditional expectation of Y given X and is a function of X . $E[\epsilon|X] = 0$ is the conditional expectation of ϵ given X and is a function of X .

Definition (Homoscedasticity noise). Fixed: $\text{Var}(y_i) = \text{Var}(\epsilon_i) = \sigma^2$ Random: $\text{Var}(Y_i | X_i) = \sigma^2$

1.2 Lecture 2: 9/3

Remark (Estimation: LS).

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - X_i^T \beta)^2$$

This is the least squares estimator for the linear regression model.

Closed-Form Solution The closed-form solution for the least squares estimator is:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

where X is the design matrix and Y is the response vector.

Remark (Estimator of σ^2). Under Homoscedasticity, the estimator of σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{n-p} \sum_{i=1}^n (Y_i - X_i^T \hat{\beta})^2$$

where p is the number of parameters in the model.

Under MLE, the estimator of σ^2 is:

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - X_i^T \hat{\beta})^2$$

where p is the number of parameters in the model. Remember that this requires a normality assumption on the noise term ϵ_i .

Proposition 1 (Homoscedastic noise in fixed design).

$$\begin{aligned} E[\hat{\beta}] &= \beta^* \\ \text{Var}(\hat{\beta}) &= (n\hat{H})^{-1} \sigma^{*2} \\ E[\hat{\sigma}^2] &= \sigma^{*2} \end{aligned}$$

Where $\hat{H} = \frac{1}{n} \sum_{i=1}^n X_i X_i^T$ is the empirical covariance matrix of the covariates X_i 's.

Remark. $E[\hat{\beta}] = \beta^*$

Proof.

$$\begin{aligned} E[\hat{\beta}] &= E[(X^T X)^{-1} X^T Y] \\ &= (X^T X)^{-1} X^T E[Y] \\ &= (X^T X)^{-1} X^T E[X\beta^* + \epsilon] \\ &= (X^T X)^{-1} X^T (X\beta^* + E[\epsilon]) \\ &= (X^T X)^{-1} X^T (X\beta^* + 0) \\ &= (X^T X)^{-1} X^T X\beta^* \\ &= \beta^* \end{aligned}$$

□

Note we only use that $E[\epsilon] = 0$ and not the distribution of ϵ

Definition (Wald Confidence Interval).

$$\hat{\beta}_j \pm z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_j)}$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the standard normal distribution. This is a confidence interval for the j -th component of the parameter vector β . For $\hat{\mu}$ the wald CI is $\hat{\mu} \pm z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\mu})}$ where $\hat{\mu}$ is the estimator of the mean μ .

Point estimator and Variance Estimator (asymptotic normality) there are things to remember for the wald CI

1.3 Lecture 3: 9/10

Remark. Remember Lin Reg: Fixed Design: $Y_i = X_i^T \beta^* + \epsilon_i, i = 1, 2, \dots, n$ where ϵ_i are independent and $E[\epsilon_i] = 0$ and $\text{Var}(\epsilon_i) = \sigma^2$.

CI for β_j^* with homoscedasticity:

$$\hat{\beta}_j \pm z_{1-\alpha/2} \hat{\sigma} \sqrt{\frac{\hat{H}_{jj}^{-1}}{n}}$$

Proposition 2 (Heteroscedastic noise).

$$E[\hat{\beta}] = \beta^*$$

$$\text{Var}(\hat{\beta}) = \hat{H}^{-1} \left(\frac{1}{n^2} \sum_i \sigma_i^{*2} X_i X_i^T \right) \hat{H}^{-1}$$

$$\text{where } \sigma_i^{*2} = \text{Var}(\epsilon_i) = \text{Var}(Y_i | X_i)$$

if σ_i^* are constant then we can pull it out and it is clear that it matches the homoscedastic case.

$$\hat{\Sigma} = \hat{H}^{-1} \frac{1}{n} \left(\sum_{i=1}^n (Y_i - X_i^T \hat{\beta})^2 X_i X_i^T \right) \hat{H}^{-1}$$

Remark.

$$\hat{\Sigma}(\beta^*) = \hat{H}^{-1} \frac{1}{n} \left(\sum_{i=1}^n (Y_i - X_i^T \beta^*)^2 X_i X_i^T \right) \hat{H}^{-1}$$

$$\sqrt{n}(\hat{\beta} - \beta^*) \rightarrow_D N(0, \hat{\Sigma}(\beta^*))$$

$$\|\hat{\Sigma}(\hat{\beta}) - \hat{\Sigma}(\beta^*)\|_F \rightarrow 0 \text{ in probability}$$

Remark (Random Design). $\hat{\beta}$ LS estimator is still good

"Conditional inference": Given $X_1, X_2, \dots, X_n$, then fixed design

"Unconditional inference": $X_1, X_2, \dots, X_n$ are random, then random design:

$$E(\hat{\beta}) = E[E[\hat{\beta} | X_i]] = E[\beta^*] = \beta^*$$

$$\text{Var}(\hat{\beta}) = E[\text{Var}(\hat{\beta} | X_i)] + \text{Var}(E[\hat{\beta} | X_i]) = E[\text{Var}(\hat{\beta} | X_i)]$$

Proposition 3.

$$\sqrt{n}(\hat{\beta} - \beta^*) \rightarrow_D N(0, \Sigma(\beta^*))$$

$$\hat{\Sigma}(\hat{\beta}^*) \rightarrow_P \Sigma(\beta^*)$$

$$\text{where } \Sigma(\beta^*) = H^{-1} G(\beta^*) H^{-1}$$

$$\text{and } H = E[X_i X_i^T]$$

$$\text{and } G(\beta^*) = E[(Y_i - X_i^T \beta^*)^2 X_i X_i^T]$$

Proof.

$$\begin{aligned}
\hat{\beta} &= \tilde{E}^{-1}[XX^T]\tilde{E}[XY] \\
\hat{\beta} - \beta^* &= \tilde{E}^{-1}[XX^T]\tilde{E}[XY] - \tilde{E}^{-1}[XX^T]\tilde{E}[XX^T]\beta^* \\
&= \tilde{E}^{-1}[XX^T]\tilde{E}[X(Y - X^T\beta^*)] \\
\sqrt{n}(\hat{\beta} - \beta^*) &= \tilde{E}^{-1}[XX^T]\sqrt{n}\tilde{E}[X(Y - X^T\beta^*)] \\
\text{By CLT, } \sqrt{n}\tilde{E}[X(Y - X^T\beta^*)] &\rightarrow_D N(0, G(\beta^*))
\end{aligned}$$

□

Where $\tilde{E}$ is the empirical expectation operator.

Theorem 1 (Slutsky's Theorem). *If $X_n \rightarrow_D X$ and $Y_n \rightarrow_P c$ then $X_n + Y_n \rightarrow_D X + c$ and $X_n Y_n \rightarrow_D cX$ and $X_n/Y_n \rightarrow_D X/c$ if $c \neq 0$*

Remark (Application of Slutsky's Theorem). We can use this to show $\text{Var}(H^{-1}Z) = H^{-1} \text{Var}(Z)H^{-1}$ and $\hat{\Sigma}(\hat{\beta}) \rightarrow_P \Sigma(\beta^*)$.

Definition (Partial Linear Regression).

$$E[Y|X] = X^T\theta^* + g^*(Z)$$

where $g(Z)$ is an unknown function of Z and Z is another covariate. This is a semi-parametric model.

Proposition 4 (1.9). *$\hat{\theta}$ is doubly robust for θ^* if either $g^*(Z)$ is correctly specified or $E[X|Z]$ is correctly specified.*

Proposition 5. *Let $V(\theta^*) = \frac{E\{\phi^2(z, w, y; \theta^*, \bar{\alpha}, \bar{\gamma})\}}{E^2\{Z(Z - W^T\bar{\gamma})\}}$ If either g or f model is correct then $\sqrt{n}(\hat{\theta} - \theta^*) \rightarrow_D N(0, V(\theta^*))$ and $\hat{V}(\hat{\theta}) \rightarrow_P V(\theta^*)$.*

Remark. $\frac{1}{n} \sum \{h(X_i; \hat{\eta}) - h(X_i; \bar{\eta})\} \rightarrow_P 0$

This says that, as $n \rightarrow \infty$, the sample-average difference between evaluating h at the estimated nuisance parameter $\hat{\eta}$ and at its target or probability limit $\bar{\eta}$ converges to zero in probability. Thus, replacing $\bar{\eta}$ with $\hat{\eta}$ has a negligible effect on the average for large samples. Equivalently, for every $\varepsilon > 0$,

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n \{h(X_i; \hat{\eta}) - h(X_i; \bar{\eta})\}\right| > \varepsilon\right) \rightarrow 0.$$

$$n^{1/2}(\hat{\eta} - \bar{\eta}) = O_P(1)$$

Definition. $\phi = (Z - W^T\bar{\gamma})(Y - W^T\bar{\alpha} - Z\theta)$

But if we choose

$$\phi_{new}(Z - \frac{\exp(W^T\gamma)}{1 + \exp(W^T\gamma)})(Y - W^T\bar{\alpha} - Z\theta)$$

then this is still mean zero and is doubly robust.